Poverty and ethnic diversity predict US metropolitan murder rates over 60 years

Luiz G. A. Alves,[1] Samvardhan Vishnoi,[2] Cássio R. A. Alves,[3] Adam R. Pah,[4] and Luís A. Nunes Amaral[1,2,5,*]

[1]*Department of Chemical and Biological Engineering, Northwestern University, Evanston, IL 60208, US*

[2]*Department of Physics and Astronomy, Northwestern University, Evanston, IL 60208, US*

[3]*Department of Economics, University of São Paulo, Ribeirão Preto, SP 14040-905, Brazil*

[4]*Department of Criminal Justice & Criminology, Georgia State University, Atlanta, GA 30303, US*

[5]*Northwestern Institute on Complex Systems (NICO), Northwestern University, Evanston, IL 60208, US*

Abstract

After nearly three decades of decline, there is again concern about the increase of the United States (US) murder rate. As forty years ago, urban areas are the hot spots. Here, we investigate the relationship between the number of murders and demographic and socio-economic metrics during the period 1960–2020. Using hierarchical regression models and Cobb–Douglas production functions, we uncover predictors of temporal and regional differences in the murder rate across 344 US metropolitan statistical areas. Specifically, we show strong evidence for urban scaling laws in cross-sectional data and that their exponents change with demographic and socio-economic conditions over time. Furthermore, we show that a Cobb–Douglas model including variables highlighted in social disorganization theory accurately captures temporal and regional fluctuations in the murder rate at the metropolitan statistical area level.

* amaral@northwestern.edu

INTRODUCTION

Violent crime is a major threat to a society's sense of well-being [1, 2]. Among developed nations, the United States (US) is an outlier with regard to the rate of violent crime and, especially, the murder rate [3–5]. The US is also anomalous in regard to the trajectory of the murder and homicide rates (Figs. 1A and S1). While most Western nations were on a sharply decreasing trend through the first half of the 20th Century, in the US it took the 1933 ratification of the 21st Amendment and the repeal of Prohibition to observe a dramatic reduction of its national homicide rate. From 1960 on, the US national homicide rate steadily increased to the levels observed in the late stages of Prohibition, and remained there for another 15 years. Unexpectedly, for much of the 1990s the national homicide rate was on a rapid decline, but this decline has become less pronounced since the turn of the millennium. The reasons for these dramatic changes have been the topic of much academic and policy discussion, but no explanation has received broad support.

In the mid-1920's, social disorganization theory (SDT) [6–8] started replacing theories ascribing innate criminality tendencies to certain races and ethnicities [9]. For early SDT scholars, neighborhood ecological characteristics replaced race and ethnicity as the cause of street level criminality. As illustrated in, e.g., Fig. 1 of Werth [10], poverty, ethnic heterogeneity, and residential instability induce the breakdown of norms and social control, and promote the cultural transmission of criminal behavior. While SDT was temporarily overtaken by psychological explanations of criminality [11–13], SDT is again being used as the justification for increased policing and 'broken-windows' policies.

SDT approaches view each distinct urban area as different because of its idiosyncratic history, ethnic breakdown, and economic conditions. In place of SDT's case study approach, some scholars have attempted to explain the dramatic rise of the national murder rate in the 1960s in terms of national level variables such as the availability of handguns [14–16], loss of economic opportunity [17, 18], increase in poverty rate [17, 19–22], demographic age-shifts [23–27], the crack-epidemic [3, 15, 28, 29], and environmental factors such as exposure to lead [2, 30]. The dramatic decrease of the murder rate in the 1990s has also been attributed to a large number of possible causes, including a dramatic increase in the incarceration rate [31], new approaches to fighting crime [30, 32–34] (some of which have been largely criticized for disproportionately targeting minorities [22, 35–37]), and improvement

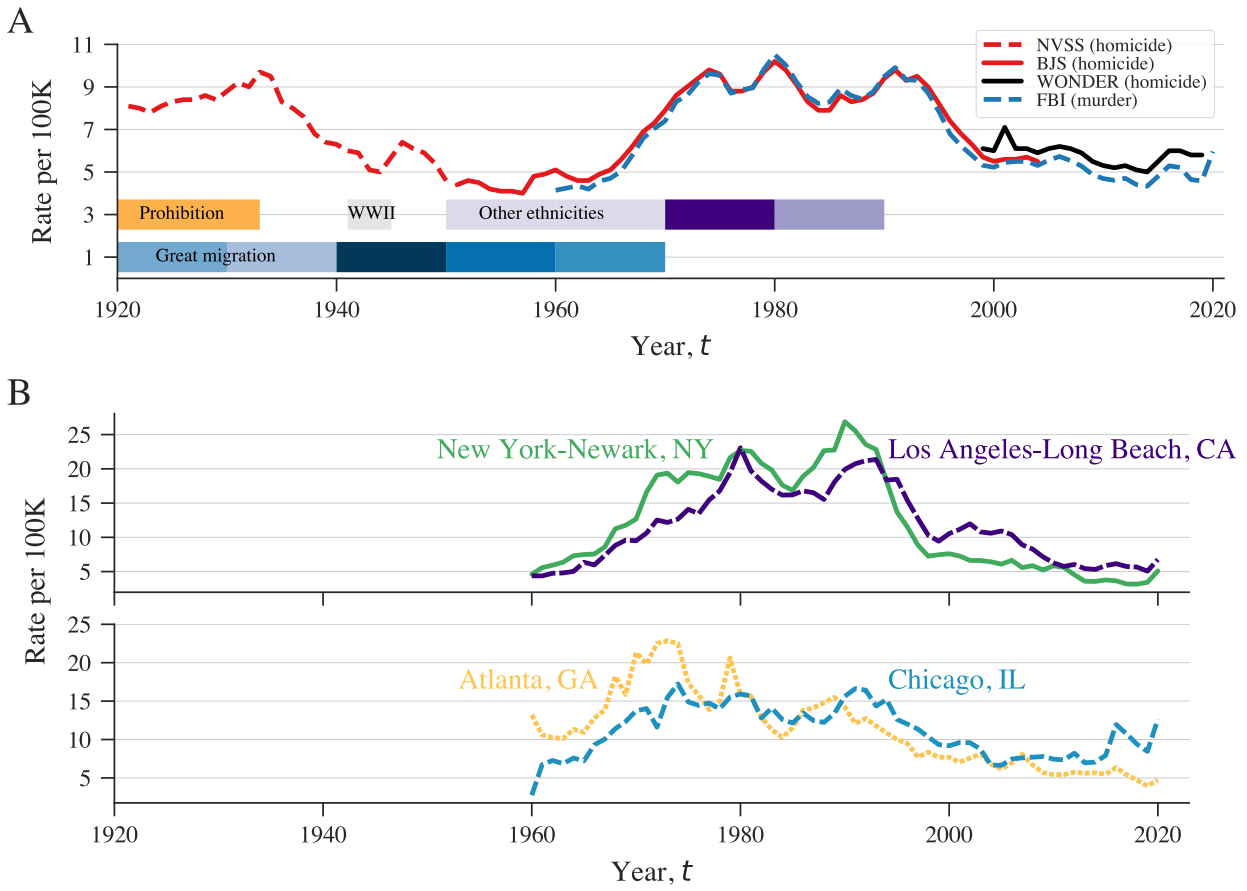


FIG. 1. **The puzzling trajectory of the US homicide rate over the past hundred years.** **(A)** Historical changes in the US Homicide rate per 100,000. We estimate the national homicide rate by combining data from four sources: Vital Statistics (1920 to 1950), Bureau of Justice Statistics (1950-2005), Federal Bureau of Investigation's Uniform Crime Reporting (2005 to 2020), and the CDC WONDER (2014 to 2020). The horizontal bars highlight significant events that could impact the national homicide rate. The blue bar displays the percentage of black and African Americans leaving Southern states. The purple bar displays the percentage increase of non-white and non-black groups within the US population. For both cases, more saturated values indicate larger percentages. The US national homicide rate is 5 to 10 times larger than that of other Western nations. **(B)** Temporal change of the murder rate for four metropolitan statistical areas (MSAs) emblematic of the Northeast, West, Midwest, and South. Note the differences in the timing and steepness of rises and falls in the murder rate for these four MSAs.

of economic and environmental aspects, and other factors [3, 38–40].

Both case studies and nationwide analyses, however, have important limitations. The former do not allow for easy generalization, whereas the latter overlooks the fact that national trends of, for example, the poverty rate, are not closely tracked at the local level (Fig. 1B). Thus, despite intense scholarly attention to this issue, we still lack a comprehensive understanding of the impact of different factors in determining the historical patterns of the US's murder rate [41, 42].

The metropolitan statistical area (MSA) offers a well-suited intermediate unit of analysis: MSAs represent economically integrated labor markets and social systems, and have been used as the canonical unit in foundational macro-level criminological research [43, 44]. Crucially, longitudinal data on poverty, ethnic composition, and crime covering six decades across hundreds of urban areas is not consistently available at finer spatial resolutions such as census tracts or neighborhoods, making the MSA the most appropriate unit for identifying the structural correlates of the long-run murder rate at national scale.

We investigate here the dynamics of the local murder rate across 344 US metropolitan statistical areas (MSAs) over the 1960-2020 period. We show that scaling theory [45] provides important insights into the factors contributing to changes in the murder rate. We also show that a Cobb-Douglas modeling approach is better suited than simple scaling to capture the fluctuations of crime over time and provides insights on the impact of demographic and socio-economic changes on local crime rates in urban areas.

RESULTS

Scaling laws in urban systems

Nearly a century ago, Cobb and Douglas [46] proposed and tested empirically a production function establishing an affine relation between outputs and inputs. Affine, power-law, relationships were further documented by Zipf [47] and have been the focus of continued research since. In the context of urban studies, researchers have documented a scaling relationship between MSA population P and the intensity Y_k of certain human activities,

$$Y_k \propto P^{\beta_k}, \tag{1}$$

for $P \gg 1$, and for Y_k representing crime rate, patent production, income, electrical cable lengths, and so on [45, 48–57]. For example, most of the *cross-sectional* variance in crime

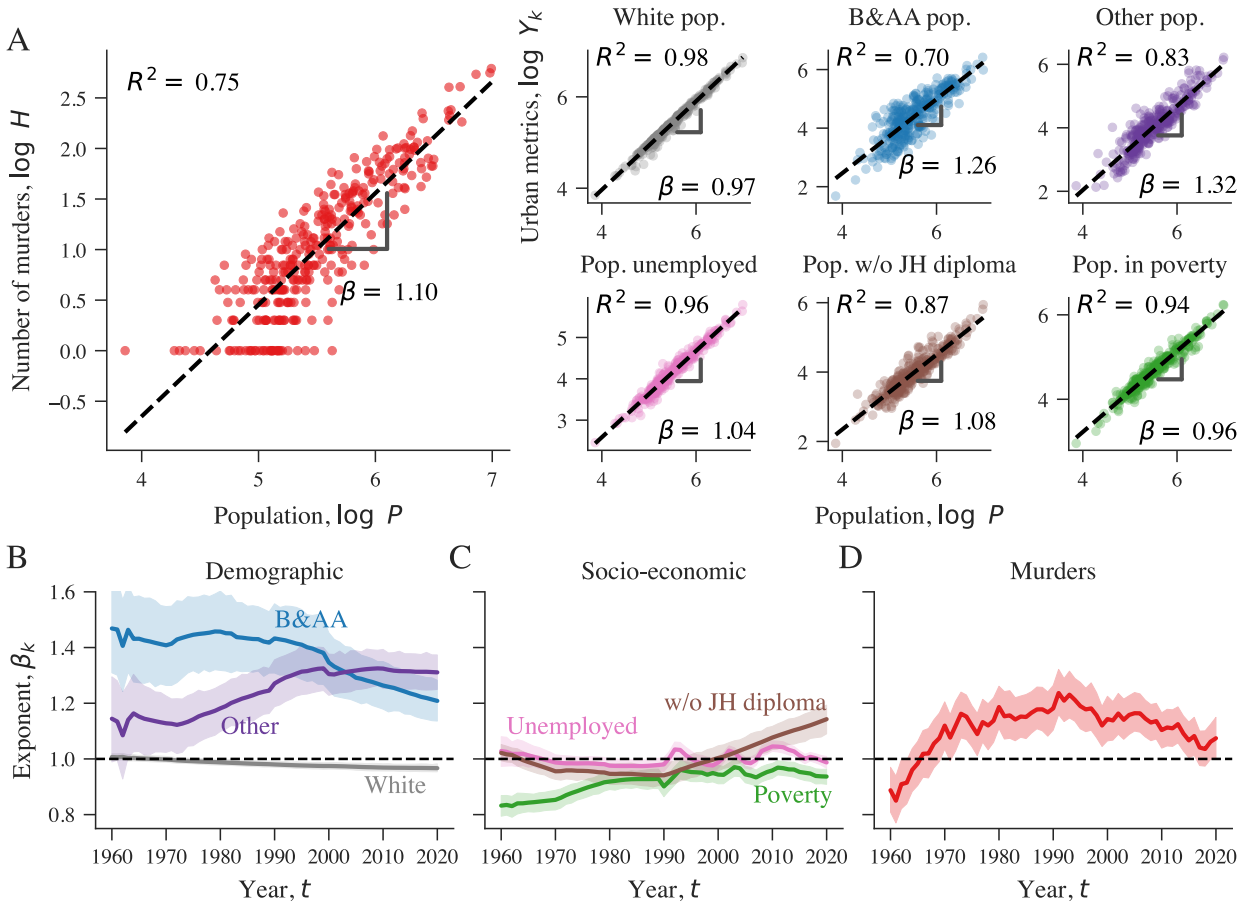


FIG. 2. **Cross-sectional variation in many human activities can be well described by parsimonious scaling laws and the changes over time in the values of the scaling exponents retrace known demographic and socio-economic changes. (A)** For illustration, we plot data for 2010 after log (10-base) transformation of the population size and the metric of interest. Similar plots can be obtained for every year in the period 1960–2020. The dashed black lines show ordinary least squares fits to the data. All scaling exponents are different from zero ($p-$values $< 10^{-14}$). **(B)** Estimates of the value of the scaling exponent for each metric (solid line) and 95% confidence intervals obtained via bootstrapping (shaded area).

rates [45, 50] can be explained by population size alone ($R^2 > 0.8$). Importantly, such scaling laws can be seen as special cases of the Cobb-Douglas production function [46, 58].

The values of the scaling exponents β_k can be used to classify the corresponding human activity in terms of how scale impacts output (Fig. 2A). $\beta > 1$ indicates increasing returns to

scale and is observed for metrics such as creation of wealth, creativity, crime, and spreading of diseases [45]. $\beta < 1$ indicates what is typically described as decreasing returns to scale and is observed for metrics such as length of electrical cabling and road surface, or number of gas stations.

Inspired by these observations, we investigate the relation between population size and other demographic variables from the perspective of scaling theory. During the period studied here, the total US population, the fraction of the population living in urban and suburban areas, and the ethnic diversity of large population centers has increased. This is visible in the fact that the White MSA population has been growing sub-linearly ($\beta < 1$) with total population, whereas the Black and African-American (B & AA) population and the population from other ethnic groups has been growing super-linearly ($\beta > 1$) with the MSA population. Suggestively, the accelerating increase in diversity of large population urban areas starts in the early 1960s and peaks in 2000 (Fig. 2B).

The scaling of socio-economic metrics with population size does not display as much variability (Fig. 2C). Whereas the scaling of the number of unemployed individuals (i.e., population unemployed) is approximately linear in the period 1960–2020, the scaling of the number of individuals with low educational attainment (i.e., population without a junior high diploma) is characterized, since 2000, by an exponent greater than 1, indicating lower educational attainment for populations residing in larger metropolitan areas. In contrast to educational attainment, population in poverty had a sub-linear relation with MSA population until 1980 indicating larger poverty rates in smaller metropolitan areas.

The story suggested by the homicides and murders data is even more intriguing (Fig. 2D). In 1960, the number of murders increased sub-linearly with population size; that is, large MSAs had *lower* murder rates than small MSAs. However, in approximately one decade, the scaling exponent grew to about 1.2, indicating a much higher murder rate for the largest MSAs. This super-linearity was maintained for the rest of the observed period, but has been declining in recent years.

Tackling the temporal patterns of the local murder rate

Despite the strong correlation between number H of homicides and murders and MSA population at the cross-sectional level, the fact remains that the temporal changes for a

given MSA are not predicted by changes in population size alone [53, 55, 56]. Indeed, because the magnitude of the temporal changes in population size is usually very large compared to that of cross-sectional data, other factors might be hidden in the deviations from the predictions of the scaling laws.

In order to address this limitation of urban scaling theory, we turn to the Cobb-Douglas formalism and the set of variables identified by SDT. SDT identifies poverty, ethnic heterogeneity, and residential instability as the primary structural correlates of urban crime [7, 8]. We operationalize these constructs using Census-derived measures consistently available across our full study period. The population below the federal poverty line captures economic disadvantage [17, 21]; counts of Black & African American residents and population from other ethnic groups capture ethnic composition and heterogeneity [8]; the population over 25 without a junior high school diploma serves as a proxy for educational disadvantage, a correlate of diminished social capital and limited economic opportunity [59]; and the unemployed population captures short-term economic dislocation [18]. Data limitations over a 60-year, 344-MSA panel preclude finer-grained constructs such as neighborhood-level residential turnover or gang activity, which lack consistent longitudinal measurement.

Specifically, we assume that

$$H \propto \Pi_{k=1}^{n} \; Y_k^{b_k} \tag{2}$$

where, for example, $Y_1 \cdots Y_g$ represent the populations of g different ethnic groups (so $\sum_{k=1}^{g} Y_k = P$), $Y_{g+1} \cdots Y_{g+e}$ represent socio-economic factors, and $Y_{g+e+1} \cdots Y_n$ represent other potentially relevant factors. If we define log-transformed variables, $h \equiv \log H$ and $y_k \equiv \log Y_k$, and make the dependence on the MSA explicit with the subscript i and its membership in some US region with the superscript $r(i)$, then we can write

$$h_{it} = a_t^{r(i)} + \sum_{k=1}^{n} \; b_{kt}^{r(i)} \; y_{kit} + \epsilon_{it} \tag{3}$$

where any time- and region-dependent intercept in Eq. (2) are absorbed by the $\left\{a_t^{r(i)}\right\}$. Note that the $\left\{b_{kt}^{r(i)}\right\}$ are potentially time and region-dependent linear coefficients, and that the $\{\epsilon_{it}\}$ are independent, normally-distributed residuals. Even though Eq. (3) is a simplified model, it includes many possible specifications with regard to which MSAs are aggregated into regions and whether coefficients do indeed depend on time and/or region. While there are many possible methods that could be used to estimate the parameters (see methods for a discussion), we chose to implement a Bayesian hierarchical regression approach [60–62]

because it allows for linear regression models with intercepts and slopes that vary over time and geographic regions (Fig. 3).

The simplest specification we consider is a *pooled model* where all parameters remain constant over region and time. The pooled model assumes that observations are independent of region and of time. Both assumptions are unlikely to be correct. The former in view of the presence of spatial inter-city correlations in crime rates [63], and the latter in view of the strong temporal persistence observed in the data (Fig. 1B). Thus, the pooled model is bound to underfit the data. Fortunately, the hierarchical approach allows us to build a set of increasingly more complex specifications allowing for time and region-varying parameters, as discussed in the Supplementary Materials.

For all specifications of Eq. (3), we consider that each parameter is drawn from a normal prior distribution $\mathbf{N}(\mu_\theta, \sigma_\theta)$, where the hyperparameters of the distribution are indexed by θ and can vary over time and region. To account for error variances that differ across observations, we consider that the variance of the model follows a generalized form of Taylor's law [64–66] (see Supplementary Material). Thus, the variance of the model can be written as, $\sigma^2 \sim \exp(\gamma + \sum_k \delta_k y_{kit})$ where k indicates the k-th urban metric, and i and t stand for the MSA and year, respectively. We assume that the standard deviation of the model residuals is drawn from a gamma prior distribution, $\sigma \sim \mathbf{\Gamma}(1/4, 1/4)$.

The hierarchical approach enables to consider specifications of Eq. (3) where the hyperparameters μ_θ and σ_θ are themselves drawn from hyperprior distributions — normal for the mean and gamma for the variance (see full description in the Supplementary Material).

For a better understanding of the robustness of our results, we consider several additional specifications of Eq. (3) — details on the estimation procedure for all different specifications can be found in the Methods and in the Supplementary Materials. A particularly interesting specification involves time-varying parameters that are modeled with random walk priors. For this specification, the parameters values for the first year in the analysis are drawn following the approach above, but for the subsequent years, the parameters are drawn from normal priors with means given by the previous year estimates.

In order to select the model with the best balance between complexity and accuracy, we use the widely applicable Bayesian information criterion (WAIC) [67, 68]. Higher WAIC scores indicate better models. Note that there is a scaling factor that can be positive or negative and it can invert the rank of models. We implement the WAIC as in arviz v0.14 [69],

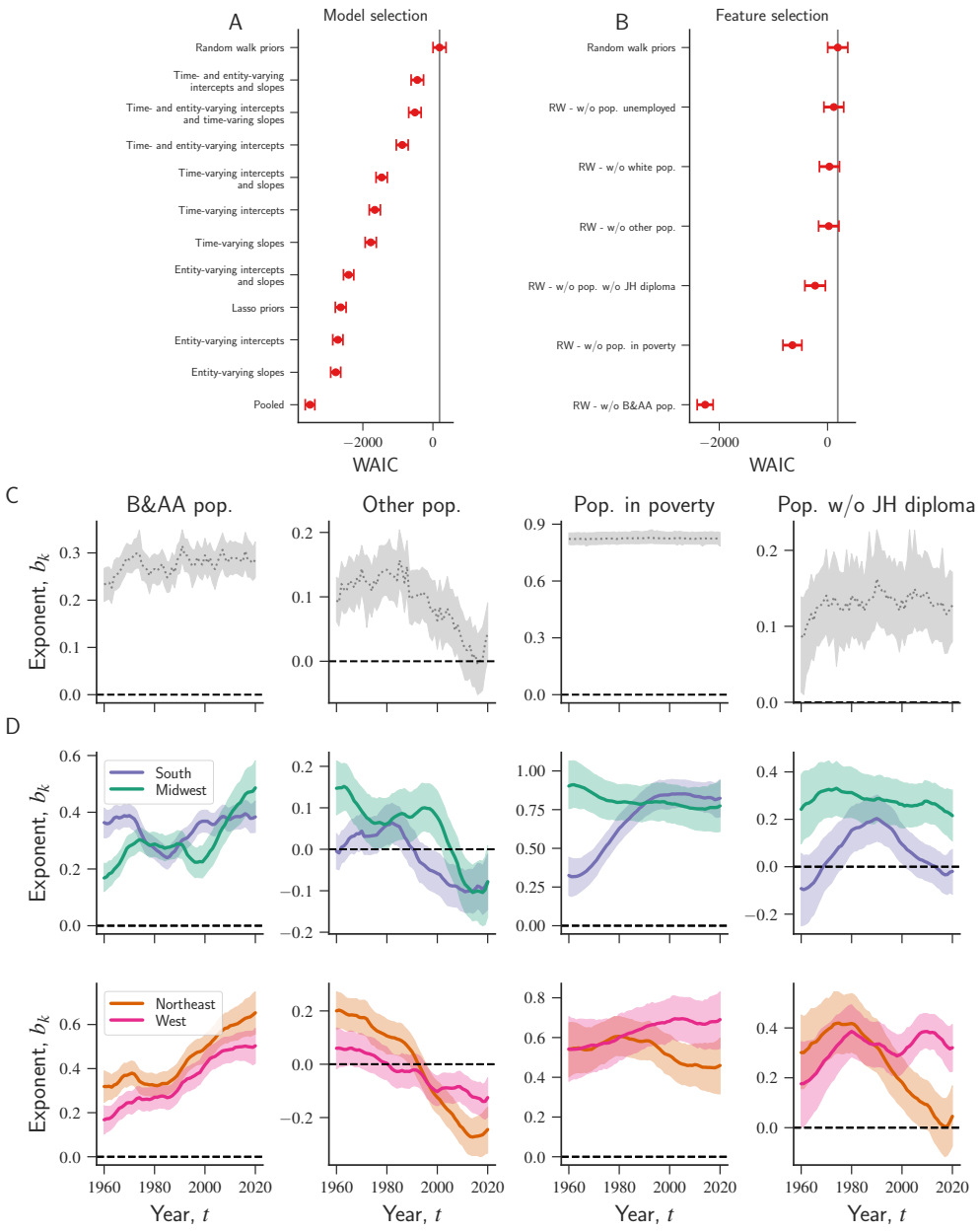


FIG. 3. **Model selection and coefficients estimates for Cobb-Douglas model.** **(A)** Widely applicable Bayesian information criterion (WAIC) score for different specifications of the Cobb-Douglas model. The highest average score is highlighted with the vertical gray line. The horizontal bars show standard error estimates. **(B)** WAIC score comparison of random walk priors model with varying feature subsets. **(C)** Bayesian estimates of the coefficients $\{b_{kit}\}$ fitting an individual model for each year. The gray dotted lines show the best estimates of the exponents and the gray shaded areas denote the 95% credible intervals. **(D)** Bayesian estimates of the coefficients $\{b_{kit}\}$ for the time- and region-varying intercepts and slopes Cobb-Douglas model, with random walk priors. The solid lines display the best estimates and the shaded regions denote 95% credible intervals.

so higher values represent better models.

Figure 3A shows that the time- and region-varying intercepts and slopes model with random walk priors yields the most accurate and parsimonious estimates for murders in urban areas. Figure 3B shows the WAIC scores of models considering subsets of features, thus unveiling the impact of each feature on the model performance. The results for this model suggest that unemployment and white population are the least predictive features—when we exclude them, we do not observe significant losses in performance.

Figure 3C shows the estimated coefficients for a specification assuming no regional effects and time-dependence of coefficients without temporal correlations for the four most relevant features. Figure 3D shows the estimated coefficients for a specification assuming random walk priors for time- and region-varying intercepts and slopes. We focus on this case given the best fit. B & AA population and population in poverty have the largest coefficients. The population in poverty coefficient is strong and stable throughout the study period, except for the South, which rises gradually from 1960 and also reaches similar levels by 1990. The coefficient for B & AA population shows a rise since around 1990 for all regions, reaching levels similar to the coefficient for population in poverty. The coefficients for population w/o JH diploma and population from other ethnic group have smaller magnitudes than the other two. The coefficient for population w/o JH diploma has been decreasing since around 1990 for both the Northeast and the South regions. The coefficient for population from other ethnic groups shows a decreasing trend since the start of the study period, reaching negative values for all regions by the year 2000.

In order to determine the extent to which the Cobb-Douglas formulation improves the accuracy of predictions over the urban scaling approach, we also use a hierarchical Bayesian approach to fit the urban scaling model to the data. As for the Cobb-Douglas model, the best hierarchical specification of the urban scaling model is one with random walk priors for time- and region-varying intercepts and slopes, shown in Figure S2. We compare the observed vs. predicted number of murders for all 344 MSAs and all years for the best specification of both the urban scaling model and Cobb-Douglas model. The performance of the Cobb-Douglas model (WAIC = 189) is significantly higher than that of the urban scaling model (WAIC = −447) with ΔWAIC = 636, indicating that the Cobb-Douglas model is significantly better descriptor of our data.

To further establish the high performance of the Cobb-Douglas model in capturing rate-

level variations, we show in Fig. 4 the predicted number of murders for four emblematic MSAs; notably, the data is always within the 95% credible intervals of the model estimates for the number of murders. Additionally, in many cases, the posterior mean estimate of the number of murders closely tracks the observed number of murders, though the 95% credible intervals reflect genuine predictive uncertainty rather than exact point predictions.

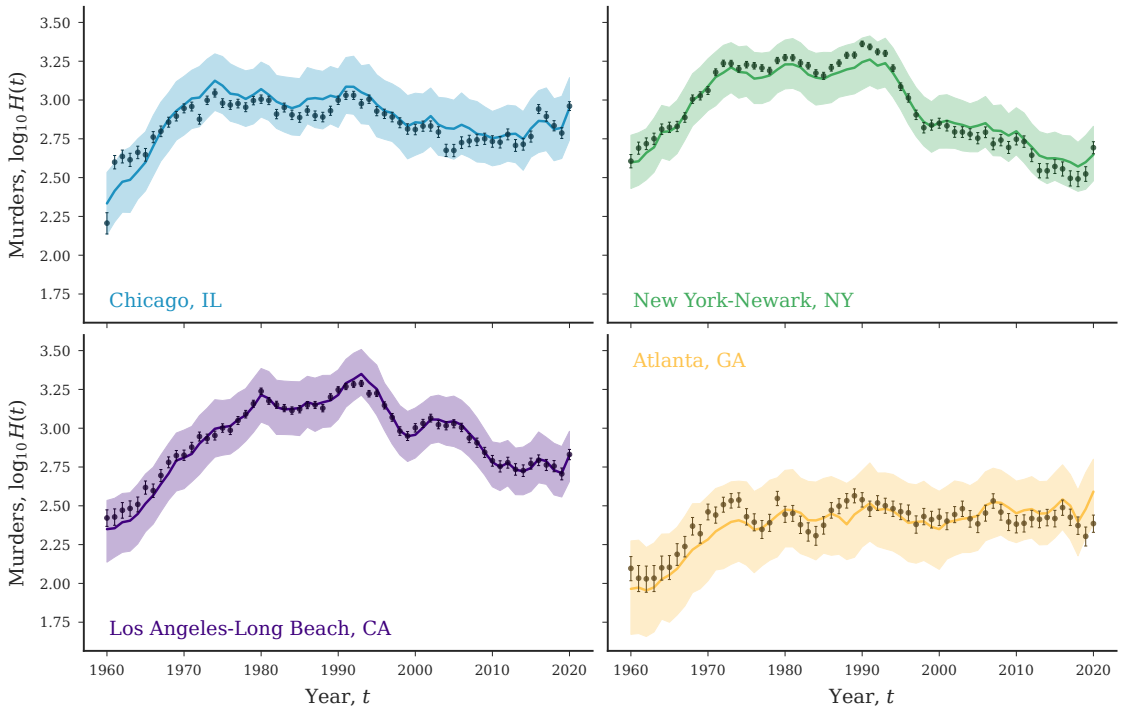


FIG. 4. **Temporal prediction considering the best Cobb-Douglas model fit for representative MSAs.** The solid line is the average estimates of number of murders, $\log H$, and the shaded area is the 95% credible intervals. Each circle is the observed number of murders for each year in the specified MSA, with 95% Poisson confidence intervals.

DISCUSSION

Many explanations have been proposed for the puzzling trajectory of the murder rate across US urban areas. Here, we demonstrated that a Cobb-Douglas formulation grounded on the ethnic diversity and socio-economic variables highlighted by SDT is able to closely capture cross-sectional variations in 344 MSA murder rates over a 60-year time period.

The strength of our model in explaining cross-sectional variations is possible due to our use

of recently-proposed model fitting approaches; specifically hierarchical Bayesian approaches that are highly-flexible in allowing for complex model specifications. Indeed, while in the past there has been considerable debate on how to address panel data with large cross-sectional effects and weaker temporal dependencies, our study shows that these challenges can be successfully tackled with these techniques.

Our model supports a strong positive association between murder rate and poverty (Fig. 3C), and the persistence of the relationship—across all four geographic regions and the full 60-year period (Fig. 3D)—carries direct policy implications. Interventions that reduce concentrated poverty in urban areas—such as housing mobility programs that relocate low-income families to lower-poverty neighborhoods [70] and investment in community-based organizations [71]—have been shown to produce measurable reductions in violent crime rates in randomized and quasi-experimental settings. Our results lend direct empirical support to this evidence, and underscore the importance of sustained anti-poverty policy as a complement to criminal justice interventions. Our model also supports a negative association between educational attainment and murder rate, albeit to a lesser extent. This is consistent with causal evidence that high school graduation reduces participation in violent crime—including murder—by raising the opportunity cost of criminal activity [72]. The existence of these associations is particularly worrisome in view of the potential impacts of the COVID-19 pandemic on both poverty (especially because the social net extended at the start of the pandemic has now been removed) and because of the negative impact of remote formats on educational attainment.

Our model also supports a strong positive association between murder rates and ethnic diversity, as predicted by SDT. That is, urban areas with a "larger than typical" number of non-white residents also experience larger murder rates. SDT suggests that this association arises from an erosion of collective efficacy—the shared norms and mutual trust that allow communities to regulate themselves informally—in ethnically diverse neighborhoods [59].

However, it is important to note that our results do not necessarily imply that SDT is the correct framework to understand this phenomenon. As Werth [10] argues, the social disharmony and racial injustice theory (SD&RIT) proposed by W.E.B. Du Bois highlights the same factors as driving the crime rate, but with a different emphasis on how those factors emerge as crime drivers. Specifically, SD&RIT highlights the role of deliberate social exclusion as a driver of both poverty and historical racial/ethnic tensions. Indeed, the

decrease in importance of the coefficient for 'Other populations' (Fig. 3C) suggests that groups other than Black and African Americans are becoming accepted as 'White,'—and thus experiencing less racial injustice—similarly to what occurred to Southern Europeans and the Irish earlier in the 20th Century.

Our work is not without limitations. First, our model does not account for variation in age structure across MSAs, particularly the proportion of the population in the criminologically high-risk 15–29 age cohort, which has been identified as a demographic predictor of the murder rate [24, 26]. Second, we have not incorporated metrics such as gun ownership, number of law enforcement officers, or prison population. For gun ownership, consistent MSA-level longitudinal data is unavailable for the full 1960-2020 study period, although state-level proxy measures on firearm suicide rates do exist [73]. Number of law enforcement officers and prison population are plausibly endogenous to the murder rate itself—jurisdictions respond to high crime with increased policing and incarceration [74, 75]—and properly identifying their causal effects would require an instrumental variable approach. It is possible that inclusion of some of these variables in future work would contribute to a greater explained-variation, or even changes in the relative importance of the different factors.

Finally, the data we analyze is far from perfect. The reporting of crime, even murder, is not consistently done by all responsible agencies and there may be under-counting — murders might be counted as disappearances when a body is not found, and murders of unregistered immigrants may not be reported to the police. These concerns may be assuaged by the fact that the unexplained variability is quite low and that the listed factors would affect different states at different times and with different magnitudes.

METHODS

Data

We retrieved data aggregated by the Federal Bureau of Investigation (FBI) at [76] and [77]. These data include number of murders at the agency level from every county in the US during the period from 1960 to 2020. Because some of the local agencies fail to report their number for some years, we used a simple outlier detection algorithm to identify such

instances and replace the values with the average of the previous and following years. Our algorithm looked for any instance where a particular observation for an agency fell below the 5th percentile of number of murders for that agency.

After imputing missing values, we aggregated the data from different agencies according to metropolitan statistical area so as to match these data with demographic and socio-economic data sets.

We retrieved demographic and socio-economic information at county level from various sources. Total and white population comprises of a dataset for 1960, that was collected using the National Historical Geographic Information System (NHGIS) [78], and the annual estimates of black, white, and total population for the periods from 1970 to 1989 and 1990 to 2019, that were obtained from the US Census Bureau [79] and the CDC WONDER database [80], respectively. The annual estimates of the population below poverty level were obtained from the US Census Bureau [79]. Poverty estimates for the years between 1960 and 1970, as well as the years 1991, 1992, 1994, and 1996 were missing from the original dataset. The estimates for Unemployed population for the years 1960 and 1970 were obtained from the NHGIS [78]; for the period spanning from 1990-2019, the annual Unemployed population was collected from the US Bureau of Labor Statistics [81]. The population over 25 years with less than a 9th grade education was collected from the NHGIS [78] for each of the census years.

We used quadratic interpolation to impute data in the missing years of our dataset and a quadratic extrapolation to impute the data for population without JH diploma and population unemployed in the period 1960–1970. The interpolation and extrapolation was made on the $\log(x+1)$-transformed variables and, later, was converted back to the original scale.

Parameter estimation approaches

Standard ordinary least squares (OLS) regression is sometimes used for parameterizing models such as Eq. (3). However, we believe this is unsuitable due to the strong co-linearity of the $\{y_{kijt}\}$, which would result in unstable parameter estimates. A more appropriate approach would be to use a regression with regularization terms, such as Ridge [82] and LASSO [83, 84] regressions. Unfortunately, proper use of regularization methods requires

the implementation of several intermediate steps including normalization of variables, cross-validation to estimate the regularizing term, and error propagation methods to recover and interpret the parameter value in its original scale [85], which can result in unreliable estimates for groups with few observations [61].

Instead of regularization methods, we chose to implement a Bayesian hierarchical regression approach [60–62]. This approach is able not only to recover the Ridge [86] and LASSO [87] estimations (given the appropriate prior distributions), but it also allows for more sophisticated linear regression models with intercepts and slopes that vary over time and geographical regions (Fig. 3).

Model Hierarchy

The simplest model one can consider is a *pooled model* where all parameters remain constant over region and time. That is, the pooled model ignores the intrinsic spatial and temporal structure of the observed data. At the other extreme, one can consider a separate cross-sectional fit for each region and year. Because of the large number of parameters, this approach is likely to over-fit the data and find biased parameter estimates.

An intermediate approach between these two extremes is to consider a hierarchical model with time- and region-varying parameters. Thus, by informing the appropriate data structure into the model, one can obtain better predictions and parameters estimates without over-fitting the data. An instance of such an intermediate approach is a model that includes time effects, but no regional dependence. The time effects can be implemented in the intercepts, in the slopes, or in both.

A second instance of such an intermediate approach is a model that allows for region dependence but no time effects. The regional-dependence can be implemented in the intercepts, in the slopes, or in both. Thus, for each region $r \equiv r(i)$, we can have $a_{rt} = a_r$ and/or $b_{krt} = b_r$, with $r \in \{\text{South, Northeast, Midwest, West}\}$.

The final instance of such an intermediate approach is a model that allows for both time effects and regional dependence. This class of models allows for time- and region-varying intercepts, time- and region-varying slopes, or a combination of both.

Fitting the Models

To fit the *cross-sectional* urban scaling laws shown in Fig. 2, we used a simple OLS regression model and bootstrapping to estimate the 95% confidence intervals. However, similar results are obtained using Bayesian regression with Gaussian priors to estimate the scaling exponents and normalization constant.

To investigate the hierarchical models shown in Fig. 3, we used a Bayesian approach [60–62]. The Bayesian models were implemented using the Python library *Pystan* version 2.19.1.1. We used a Markov chain Monte Carlo (MCMC) sampling algorithm, called No-U-Turn Sampler (NUTS) [88], for the statistical inference of the models parameters. To speed up the convergence of the Markov Chains, we centered the populations by subtracting the mean. We used 6,000 steps for the warm-up stage and 1,000 steps to estimate the posterior distributions of the model parameters. Using the centered data, we found that 7,000 steps were enough to guarantee convergence.

We determined convergence by computing $\hat{R}$ for each parameter being fitted. This quantity measures the convergence of different MCMC simulations, using 2-chain simulations [60]. For all models fitted, we obtained $\hat{R} \approx 1$ for all parameters, indicating that the different chains had converged [60]. Further details of the models priors and hyperpriors are provided in the Supplementary Material.

ACKNOWLEDGMENTS

LANA thanks the John and Leslie McQuown Gift. LGAA thanks Diego Fregolente for help with the FBI crime data set.

DATA AVAILABILITY

The processed data necessary to reproduce the reported analyses are available to editors and reviewers upon request. The underlying data were obtained from the Federal Bureau of Investigation Uniform Crime Reporting Program, the Inter-university Consortium for Political and Social Research, IPUMS National Historical Geographic Information System, the US Census Bureau, CDC WONDER, and the US Bureau of Labor Statistics. The

processed data and associated documentation will be deposited in a public repository upon publication.

CODE AVAILABILITY

The custom code and computational notebooks supporting this study are available to editors and reviewers upon request. The version corresponding to the published manuscript will be deposited in a public, DOI-minting repository upon publication.

---

[1] Etienne G Krug, James A Mercy, Linda L Dahlberg, and Anthony B Zwi, "The world report on violence and health," The Lancet **360**, 1083–1088 (2002).

[2] Solomon M Hsiang, Marshall Burke, and Edward Miguel, "Quantifying the influence of climate on human conflict," Science **341**, 1235367 (2013).

[3] Alfred Blumstein, Frederick P Rivara, and Richard Rosenfeld, "The rise and decline of homicide—and why," Annual Review of Public Health **21**, 505–541 (2000).

[4] Sherry Towers, Andres Gomez-Lievano, Maryam Khan, Anuj Mubayi, and Carlos Castillo-Chavez, "Contagion in mass killings and school shootings," PLoS ONE **10**, e0117259 (2015).

[5] AR Pah, J Hagan, AL Jennings, A Jain, K Albrecht, AJ Hockenberry, and LAN Amaral, "Economic insecurity and the rise in gun violence at us schools," Nature Human Behaviour **1**, 1–6 (2017).

[6] Robert E Park and Ernest W Burgess, *The city* (University of Chicago Press, 1925).

[7] Clifford Robe Shaw and Henry Donald McKay, "Juvenile delinquency and urban areas." (1942).

[8] Robert J Sampson, William Julius Wilson, John Hagan, and Ruth D Peterson, *Toward a Theory of Race, Crime, and Urban Inequality* (Stanford University Press, 1995) pp. 37–54.

[9] Francis Galton, *Hereditary genius* (D. Appleton, 1891).

[10] S Rose Werth, "Social disharmony and racial injustice: Web du bois's theories on crime," Social Problems (2022).

[11] Maurice Philip Feldman, *Criminal Behaviour: A Psychological Analysis* (Wiley London, 1977).

[12] David P Farrington, *Psychological Explanations of Crime* (Dartmouth Aldershot, UK, 1994).

[13] Christopher W Mullins, Richard Wright, and Bruce A Jacobs, "Gender, streetlife and criminal retaliation," Criminology **42**, 911–940 (2004).

[14] Alfred Blumstein, "Youth violence, guns, and the illicit-drug industry," J. Crim. L. & Criminology **86**, 10 (1995).

[15] Daniel Cork, "Examining space–time interaction in city-level homicide data: crack markets and the diffusion of guns among youth," Journal of Quantitative Criminology **15**, 379–406 (1999).

[16] Mona A Wright, Garen J Wintemute, and Frederick P Rivara, "Effectiveness of denial of handgun purchase to persons believed to be at high risk for firearm violence." American Journal of Public Health **89**, 88–90 (1999).

[17] Gary S Becker, "Crime and punishment: An economic approach," in *The Economic Dimensions of Crime* (Springer, 1968) pp. 13–68.

[18] Jeff Grogger, "Market wages and youth crime," Journal of Labor Economics **16**, 756–791 (1998).

[19] William C Bailey, "Poverty, inequality, and city homicide rates: Some not so unexpected findings," Criminology **22**, 531–550 (1984).

[20] Steven F Messner, "Poverty, inequality, and the urban homicide rate: Some unexpected findings," Criminology **20**, 103–114 (1982).

[21] Morgan Kelly, "Inequality and crime," Review of Economics and Statistics **82**, 530–539 (2000).

[22] Mike Males, "Age, poverty, homicide, and gun homicide: is young age or poverty level the key issue?" Sage Open **5**, 2158244015573359 (2015).

[23] Theodore N Ferdinand, "Demographic shifts and criminality: An inquiry," The British Journal of Criminology **10**, 169–175 (1970).

[24] Alfred Blumstein and Daniel S Nagin, "Analysis of arrest rates for trends in criminality," Socio-economic Planning Sciences **9**, 221–227 (1975).

[25] Roland Chilton, "Twenty years of homicide and robbery in Chicago: The impact of the city's changing racial and age composition," Journal of Quantitative Criminology **3**, 195–214 (1987).

[26] James Alan Fox and Alex R Piquero, "Deadly demographics: Population characteristics and forecasting homicide trends," Crime & Delinquency **49**, 339–359 (2003).

[27] Darrell Steffensmeier, Jeffery Ulmer, and John Kramer, "The interaction of race, gender, and age in criminal sentencing: The punishment cost of being young, black, and male," Criminology **36**, 763–798 (1998).

[28] Jeff Grogger and Michael Willis, "The emergence of crack cocaine and the rise in urban crime rates," Review of Economics and Statistics **82**, 519–529 (2000).

[29] James Alan Fox, "Demographics and US homicide," in *The Crime Drop in America*, edited by Alfred Blumstein and Joel Wallman (Cambridge: Cambridge University Press, 2000) pp. 288–317.

[30] Kees Keizer, Siegwart Lindenberg, and Linda Steg, "The spreading of disorder," Science **322**, 1681–1685 (2008).

[31] William Spelman, *Criminal Incapacitation* (Springer Science & Business Media, 1993).

[32] David Weisburd and John E Eck, "What can police do to reduce crime, disorder, and fear?" The Annals of the American Academy of Political and Social Science **593**, 42–65 (2004).

[33] David Weisburd, Stephen D Mastrofski, Ann Marie McNally, Rosann Greenspan, and James J Willis, "Reforming to preserve: Compstat and strategic problem solving in american policing," Criminology & Public Policy **2**, 421–456 (2003).

[34] Richard Rosenfeld, Robert Fornango, and Eric Baumer, "Did ceasefire, compstat, and exile reduce homicide?" Criminology & Public Policy **4**, 419–449 (2005).

[35] Julia Dressel and Hany Farid, "The accuracy, fairness, and limits of predicting recidivism," Science Advances **4**, eaao5580 (2018).

[36] Elizabeth Brown and Mike Males, "Does age or poverty level best predict criminal arrest and homicide rates? a preliminary investigation," Justice Policy Journal **8**, 1–30 (2011).

[37] Mike A Males and Elizabeth A Brown, "Teenagers' high arrest rates: Features of young age or youth poverty?" Journal of Adolescent Research **29**, 3–24 (2014).

[38] John J Donohue III and Steven D Levitt, "The impact of legalized abortion on crime," The Quarterly Journal of Economics **116**, 379–420 (2001).

[39] Christopher L Foote and Christopher F Goetz, "The impact of legalized abortion on crime: Comment," The Quarterly Journal of Economics **123**, 407–423 (2008).

[40] Ted Joyce, "A simple test of abortion and crime," The Review of Economics and Statistics **91**, 112–123 (2009).

[41] Alfred Blumstein, Joel Wallman, and David Farrington, *The Crime Drop in America* (Cambridge University Press, 2006).

[42] Mirta B Gordon, "A random walk in the literature on criminality: A partial and critical view on some statistical analyses and modelling approaches," European Journal of Applied Mathematics **21**, 283–306 (2010).

[43] Kenneth C. Land, Patricia L. McCall, and Lawrence E. Cohen, "Structural covariates of homicide rates: Are there any invariances across time and social space?" American Journal of Sociology **95**, 922–963 (1990).

[44] Travis C. Pratt and Francis T. Cullen, "Assessing macro-level predictors and theories of crime: A meta-analysis," Crime and Justice **32**, 373–450 (2005).

[45] Luís MA Bettencourt, José Lobo, Dirk Helbing, Christian Kühnert, and Geoffrey B West, "Growth, innovation, scaling, and the pace of life in cities," Proceedings of the National Academy of Sciences **104**, 7301–7306 (2007).

[46] Charles W Cobb and Paul H Douglas, "A theory of production," The American Economic Review **18**, 139–165 (1928).

[47] George Kingsley Zipf, *Human Behaviour and the Principle of Least Effort* (Addison-Wesley Press, 1949).

[48] Luís MA Bettencourt, José Lobo, Deborah Strumsky, and Geoffrey B West, "Urban scaling and its deviations: Revealing the structure of wealth, innovation and crime across cities," PLoS ONE **5** (2010).

[49] Luiz G A Alves, Haroldo V Ribeiro, and Renio S Mendes, "Scaling laws in the dynamics of crime growth rate," Physica A: Statistical Mechanics and its Applications **392**, 2672–2679 (2013).

[50] Luiz G A Alves, Haroldo V Ribeiro, Ervin K Lenzi, and Renio S Mendes, "Distance to the scaling law: a useful approach for unveiling relationships between crime and urban metrics," PLoS ONE **8** (2013).

[51] Luiz G A Alves, Renio S Mendes, Ervin K Lenzi, and Haroldo V Ribeiro, "Scale-adjusted metrics for predicting the evolution of urban indicators and quantifying the performance of cities," PLoS ONE **10** (2015).

[52] Andres Gomez-Lievano, Hyejin Youn, and Luis MA Bettencourt, "The statistics of urban scaling and their connection to zipf's law," PLoS ONE **7** (2012).

[53] Jules Depersin and Marc Barthelemy, "From global scaling to the dynamics of individual cities," Proceedings of the National Academy of Sciences **115**, 2317–2322 (2018).

[54] V Chuqiao Yang, Andrew V Papachristos, and Daniel M Abrams, "Modeling the origin of urban-output scaling laws," Physical Review E **100**, 032306 (2019).

[55] Marc Keuschnigg, "Scaling trajectories of cities," Proceedings of the National Academy of Sciences **116**, 13759–13761 (2019).

[56] Marc Barthelemy, "The statistical physics of cities," Nature Reviews Physics **1**, 406–415 (2019).

[57] Luís MA Bettencourt, Vicky Chuqiao Yang, José Lobo, Christopher P Kempes, Diego Rybski, and Marcus J Hamilton, "The interpretation of urban scaling analysis in time," Journal of the Royal Society Interface **17**, 20190846 (2020).

[58] David F Heathfield, *An Introduction to Cost and Production Functions* (Macmillan International Higher Education, 2016).

[59] Robert J Sampson, Stephen W Raudenbush, and Felton Earls, "Neighborhoods and violent crime: A multilevel study of collective efficacy," Science **277**, 918–924 (1997).

[60] Andrew Gelman, John B Carlin, Hal S Stern, David B Dunson, Aki Vehtari, and Donald B Rubin, *Bayesian Data Analysis* (CRC press, 2013).

[61] Andrew Gelman and Jennifer Hill, *Data Analysis Using Regression and Multilevel/Hierarchical Models* (Cambridge university press, 2006).

[62] Gary M Koop, *Bayesian Econometrics* (John Wiley & Sons Inc., 2003).

[63] Luiz G A Alves, Ervin K Lenzi, Renio S Mendes, and Haroldo V Ribeiro, "Spatial correlations, clustering and percolation-like transitions in homicide crimes," EPL (Europhysics Letters) **111**, 18002 (2015).

[64] Lionel Roy Taylor, "Aggregation, variance and the mean," Nature **189**, 732–735 (1961).

[65] Quentin S Hanley, Suniya Khatun, Amal Yosef, and Rachel-May Dyer, "Fluctuation scaling, taylor's law, and crime," PLoS One **9**, e109004 (2014).

[66] Jorge C Leitao, José Maria Miotto, Martin Gerlach, and Eduardo G Altmann, "Is this scaling nonlinear?" Royal Society open science **3**, 150649 (2016).

[67] Sumio Watanabe, "A Widely Applicable Bayesian Information Criterion," Journal of Machine Learning Research **14**, 867–897 (2013).

[68] Andrew Gelman, Jessica Hwang, and Aki Vehtari, "Understanding predictive information criteria for Bayesian models," Statistics and Computing **24**, 997–1016 (2014).

[69] "arviz.compare — arviz 0.14.0 documentation," .

[70] Raj Chetty, Nathaniel Hendren, and Lawrence F. Katz, "The effects of exposure to better neighborhoods on children: New evidence from the moving to opportunity experiment," American Economic Review **106**, 855–902 (2016).

[71] Patrick Sharkey, Gerard Torrats-Espinosa, and Delaram Takyar, "Community and the crime decline: The causal effect of local nonprofits on violent crime," American Sociological Review **82**, 1214–1240 (2017).

[72] Lance Lochner and Enrico Moretti, "The effect of education on crime: Evidence from prison inmates, arrests, and self-reports," American Economic Review **94**, 155–189 (2004).

[73] Deborah Azrael, Philip J. Cook, and Matthew Miller, "State and local prevalence of firearms ownership measurement, structure, and trends," Journal of Quantitative Criminology **20**, 43–62 (2004).

[74] Steven D. Levitt, "Using electoral cycles in police hiring to estimate the effect of police on crime," American Economic Review **87**, 270–290 (1997).

[75] Steven D. Levitt, "The effect of prison population size on crime rates: Evidence from prison overcrowding litigation," The Quarterly Journal of Economics **111**, 319–351 (1996).

[76] "FBI," http://www.ucrdatatool.gov/, accessed: 2022-01-01.

[77] "United States. Federal Bureau of Investigation. Uniform Crime Reporting Program Data: Offenses Known and Clearances by Arrest, United States, 2015-2020. Inter-university Consortium for Political and Social Research [distributor]," https://doi.org/10.3886/ICPSR37911.v1, accessed: 2022-10-05.

[78] "IPUMS NHGIS," https://www.nhgis.org/, accessed: 2022-01-01.

[79] "US Census Bureau," https://www.census.gov, accessed: 2022-01-01.

[80] "CDC WONDER: Bridged-Race Population Estimates 1990-2020," https://wonder.cdc.gov/Bridged-Race-v2020.HTML, accessed: 2022-01-01.

[81] "US Bureau of Labor Statistics," https://www.bls.gov, accessed: 2022-01-01.

[82] Arthur E Hoerl and Robert W Kennard, "Ridge regression: Biased estimation for nonorthogonal problems," Technometrics **12**, 55–67 (1970).

[83] Robert Tibshirani, "Regression shrinkage and selection via the LASSO," Journal of the Royal Statistical Society: Series B (Methodological) **58**, 267–288 (1996).

[84] Robert Tibshirani, "Regression shrinkage and selection via the LASSO: a retrospective," Journal of the Royal Statistical Society: Series B (Statistical Methodology) **73**, 273–282 (2011).

[85] Jerome Friedman, Trevor Hastie, and Robert Tibshirani, *The Elements of Statistical Learning*, Vol. 1 (Springer series in statistics New York, 2001).

[86] M Goldstein, "Bayesian analysis of regression problems," Biometrika **63**, 51–58 (1976).

[87] Chris Hans, "Bayesian LASSO regression," Biometrika **96**, 835–845 (2009).

[88] Matthew D Hoffman and Andrew Gelman, "The No-U-Turn sampler: adaptively setting path lengths in Hamiltonian Monte Carlo," J. Mach. Learn. Res. **15**, 1593–1623 (2014).

[89] Bastian Herre, Fiona Spooner, and Max Roser, "Homicides," Our World in Data (2013), https://ourworldindata.org/homicides.

[90] Herbert A Simon, "On a class of skew distribution functions," Biometrika **42**, 425–440 (1955).

[91] Benoit B Mandelbrot, *Fractals and Scaling in Finance: Discontinuity, Concentration, Risk. Selecta Volume E* (Springer Science & Business Media, 2013).

[92] Luis MA Bettencourt, "Urban growth and the emergent statistics of cities," Science Advances **6**, eaat8812 (2020).

[93] Robert K Merton, "Social structure and anomie," American Sociological Review **3**, 672–682 (1938).

[94] E Britt Patterson, "Poverty, income inequality, and community crime rates," Criminology **29**, 755–776 (1991).